Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Computer Science and Engineering

Lab Report

For the course

CSE 3206: Software Engineering Sessional

By
MD Rakibul Hasan Rifat
2103057

To
Farjana Parvin
Assistant Professor,
Dept. Of CSE, RUET

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Task: **Software Configuration Management with Git**

Overview:

A GitHub repository was established as a collaborative development project demonstrating version control practices and teamwork. The project involved three contributors working together using Git and GitHub workflows.

Git/GitHub Functions Performed:

Repository Initialization

- **Repository Creation**: Established the base repository structure
- **Initial Commit**: First commit created the project foundation

Branch Creation: Three branches were created

- main - Primary development branch
- master - Secondary integration branch
- 2103033_Johan - Feature branch for calculator implementation

Forking

- **Fork #1**: heytamjid created fork calcfork for independent development
- **Fork #2**: Asif-Johan created fork cse-3206_test_fork for testing contributions

Pull Requests

- **PR #1**: Merged Johan's calculator implementation from 2103033_Johan branch
- **PR #2**: Merged Rifat's contributions from master branch
- **PR #3**: Opened by Asif-Johan from his fork (contributed layout fixes)
- **PR #4**: Merged Tamjid's modified calculator from fork branch fb

Merge Operations

- Successfully merged 4 pull requests into main branch
- Performed branch-to-branch merge operations
- Integrated changes from multiple forks

Collaboration Management

- **Collaborator Additions**: Two team members added as collaborators

- **Code Reviews**: Changes reviewed through pull request workflow
- **Conflict Resolution**: Managed merge conflicts during integration

Configuration Management

- **.gitignore Implementation**: Configured to exclude unnecessary files
- **Version Control**: Maintained commit history with descriptive messages

Git Commands

git config --global user.name "MohammadRakibulHasan04"

git config --globaluser.email rakibulhasanrifat04@gmail.com

git init

git add .

git commit -m "Initial commit"

git branch -M main

git remote add origin https://github.com/MohammadRakibulHasan04/cse-3206

git push -u origin main

git checkout main

git pull origin main

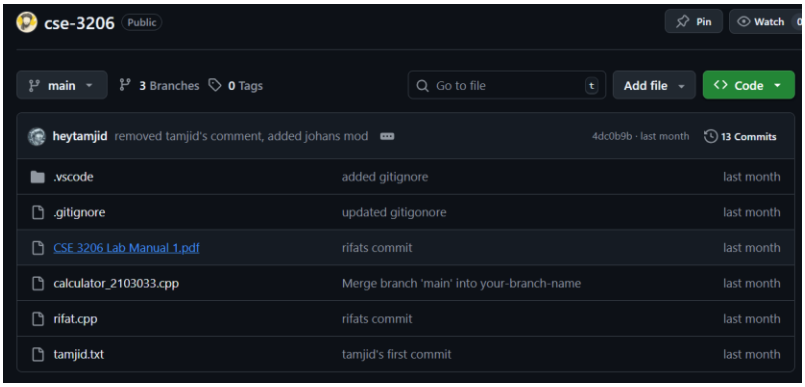
git checkout -b master

git status

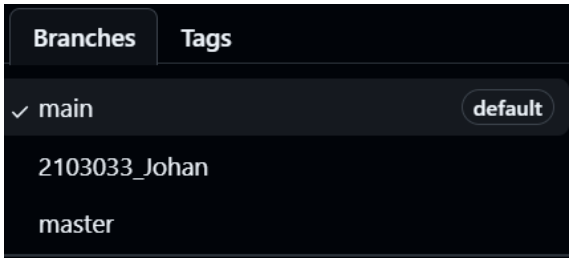
git add .

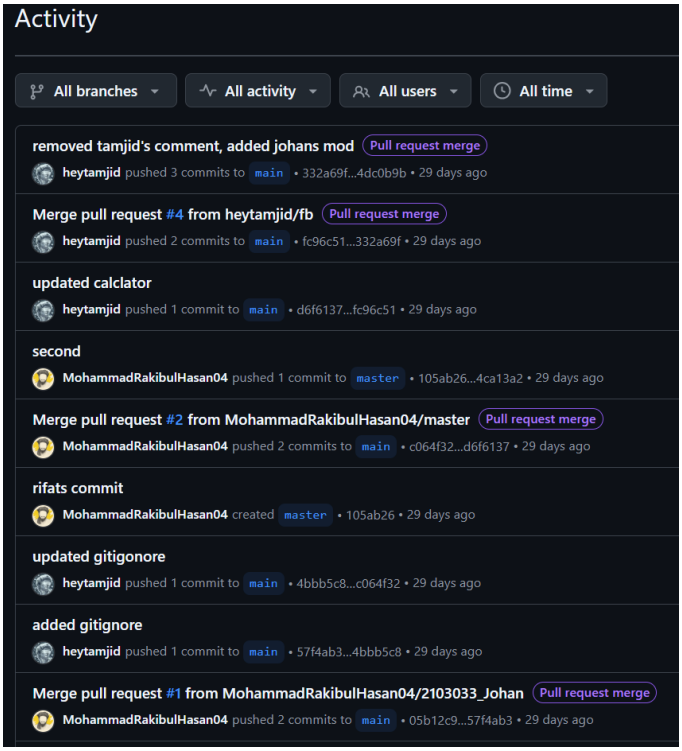
git commit -m "rifats commit"

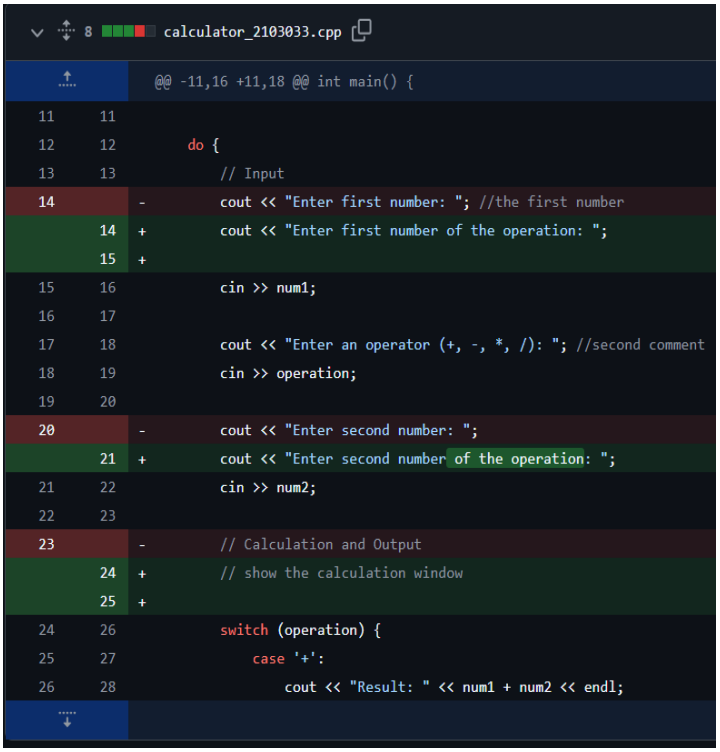
Results:


cse-3206
Public
Pin
Watch
main
3 Branches
0 Tags
Go to file
Add file
Code
heytamjid removed tamjid's comment, added johans mod
4dc0b9b · last month
13 Commits
.vscode added gitignore last month
.gitignore updated gitigonore last month
CSE 3206 Lab Manual 1.pdf rifats commit last month
calculator_2103033.cpp Merge branch 'main' into your-branch-name last month
rifat.cpp rifats commit last month
tamjid.txt tamjid's first commit last month


Contributors 3
heytamjid TAMJID HASAN FAHIM
MohammadRakibulHasan04 MD. RAKI...
Asif-Johan Asif Johan


Branches
Tags
main
default
2103033_Johan
master


Activity
All branches
All activity
All users
All time
removed tamjid's comment, added johans mod Pull request merge
heytamjid pushed 3 commits to main • 332a69f...4dc0b9b • 29 days ago
Merge pull request #4 from heytamjid/fb Pull request merge
heytamjid pushed 2 commits to main • fc96c51...332a69f • 29 days ago
updated calclator
heytamjid pushed 1 commit to main • d6f6137...fc96c51 • 29 days ago
second
MohammadRakibulHasan04 pushed 1 commit to master • 105ab26...4ca13a2 • 29 days ago
Merge pull request #2 from MohammadRakibulHasan04/master Pull request merge
MohammadRakibulHasan04 pushed 2 commits to main • c064f32...d6f6137 • 29 days ago
rifats commit
MohammadRakibulHasan04 created master • 105ab26 • 29 days ago
updated gitigonore
heytamjid pushed 1 commit to main • 4bbb5c8...c064f32 • 29 days ago
added gitignore
heytamjid pushed 1 commit to main • 57f4ab3...4bbb5c8 • 29 days ago
Merge pull request #1 from MohammadRakibulHasan04/2103033_Johan Pull request merge
MohammadRakibulHasan04 pushed 2 commits to main • 05b12c9...57f4ab3 • 29 days ago


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calculator_2103033.cpp
@@ -11,16 +11,18 @@ int main() {
do {
// Input
- cout << "Enter first number: "; //the first number
+ cout << "Enter first number of the operation: ";
+
cin >> num1;
cout << "Enter an operator (+, -, *, /): "; //second comment
cin >> operation;
- cout << "Enter second number: ";
+ cout << "Enter second number of the operation: ";
cin >> num2;
- // Calculation and Output
+ // show the calculation window
+
switch (operation) {
case '+':
cout << "Result: " << num1 + num2 << endl;

Conclusion:

Core Git/GitHub workflows—initialization, branching, forking, PRs, and merges were applied in this repository. Some problems were introduced which were solved then.